

Building AI Agents for Procedures

Project Overview

Construction workers often come across unexpected events out in the field, for which they need to bring in outside expertise. Usually, the first escalation is to a team leader or manager, but they are sometimes unavailable. Technicians can also turn to their company-provided procedures, but these are often verbose and difficult to navigate quickly.

AI Agents could be a viable solution to this. AI agents are custom-built using an agent builder and designed to handle tasks surrounding specific topics. A common use case is to leverage NLP to answer questions without external input.

This project aims to build out AI Agents around generic, but wide-spanning procedures online and demonstrate how reliable they are when answering questions about those procedures. Students should create metrics on evaluating the reliability of the AI agent, create recorded demonstrations for the purposes of training, and extensively document the creation and training process.

Deliverables

A successful completion of this project will produce:

- A created AI agent that can answer nearly any question about the procedures its trained on.
- A repository of questions that the agent cannot answer to identify gaps in existing procedures.
- Video demonstrations of students interacting with the agent to demonstrate its function.
- A risk assessment and cost-benefit analysis of its implementation within a company.
- Documentation of the agent's creation, training, tuning, testing, and updating.